

Quantitative Aptitude

Ratio & Proportions

Level-1

Q1 If $p = \frac{1}{3}q$ and $q = \frac{1}{2}r$, then $p:q:r$, is equal to ?

- (A) 1:3:1 (B) 1:3:6
(C) 3:1:2 (D) 2:1:3
(E) None of these

Q2 If 30% of $(a + b) = 40\%$ of $(a - b)$, then find $a : b$

- (A) 7:1 (B) 1:7
(C) 3:7 (D) 1:2
(E) None of these

Q3 Find the fourth Proportion of 36, 48, and 72.

- (A) 84 (B) 96
(C) 18 (D) 24
(E) None of these

Q4 If A is twice of B and C is twice of B then the ratio of A:B: C is ?

- (A) 1 : 2 : 2 (B) 1 : 1 : 2
(C) 2 : 3 : 4 (D) 1 : 2 : 3
(E) 2 : 1 : 2

Q5 If $4A = 5B = 6C$ then find the ratio of A : B : C.

- (A) 12 : 15 : 10 (B) 10 : 12 : 15
(C) 15 : 10 : 12 (D) 15 : 12 : 10
(E) 16 : 10 : 11

Q6 If $(5x + 2y) : (4x + 9y) = 21 : 13$, What is the value of $x : y$?

- (A) 19:163 (B) 163:19
(C) 19:63 (D) 19:47
(E) None of these

Q7 The ratio of the number of red balls to blue balls in a bag is 3:5. If there are total 24 balls in the bag, how many red balls are there?

- (A) 15 balls (B) 13 balls
(C) 9 balls (D) 10 balls
(E) 12 balls

Q8

If a particular number is subtracted from each of 17, 23, 27, and 38, the results of numbers are in proportion. Find the number to be subtracted.

- (A) 1 (B) 2
(C) 3 (D) 4
(E) 5

Q9 The salary of Suchi, Anita and Roshni are in ratio 5:7:9, respectively. If the salary of Roshni is Rs. 6,300, then find the sum (in Rs.) of salaries of Suchi, Anita and Roshni together.

- (A) 14700 (B) 15700
(C) 16700 (D) 13700
(E) None of these

Q10 The ratio of first to second number is 2:3 and ratio of second to third number is 5:8. Sum of all three numbers is 1176. Find the third number.

- (A) 434 (B) 322
(C) 576 (D) 400
(E) 350

Q11 The flight fare between two cities is increased in the ratio 9:11. What is the increase (in Rs) in the fare, if the original fare was Rs.18,000?

- (A) 8800 (B) 5600
(C) 4000 (D) 9900
(E) None of these

Q12 The ratio of monthly income of Riya and Supriya is 5:7 and that of their monthly expenditure is 7:10. If the income of Riya is equal to expenditure of Supriya then what is the ratio of saving of Riya and Supriya?

- (A) 2:3 (B) 4:3
(C) 3:4 (D) 3:2
(E) 5:3

Q13



The ratio of the price of a bed and a table is 9 : 7. The average price of two beds and a table is Rs. 60,000. The total sum of the price of a bed and a table is ?

- (A) Rs. 1,51,200 (B) Rs. 1,15,000
(C) Rs. 1,15,200 (D) Rs. 1,50,000
(E) None of these

Q14 The ratio of three numbers is 3:5:9 and sum of their squares is 4140. Find the sum of the numbers.

- (A) 112 (B) 100
(C) 102 (D) 92
(E) None of these

Q15 The total number of students in a school is 1680 in which the number of girls is 360 less than the number of boys. Find the respective ratio between the number of boys and the number of girls.

- (A) 17 : 21
(B) 17 : 12
(C) 17 : 11
(D) 13 : 11
(E) None of these

Q16 The ratio of three numbers A, B and C is 13:17:23 and their sum is 265. Find the ratio between the sum of A and B to the difference of C and B?

- (A) 5:1 (B) 1:5
(C) 2:5 (D) 3:2
(E) None of these

Q17 The ratio of the incomes of M to N is 8:11. If each saves Rs. 6000, the ratio of expenditure of M to N becomes 5:8. Find the income of M.

- (A) Rs. 16000 (B) Rs. 18000
(C) Rs. 10000 (D) Rs. 12000
(E) None of these

Q18 The ratio of the income of Utkarsh and Radhika is 4:3 and the ratio of their expenditures is 3:2. If at the end of the year, each saves Rs 2500, then the respective income of each will be:

- (A) Rs 10000 , Rs 7500
(B) Rs 9500, Rs 7000

(C) Rs 7500, Rs 10000

(D) Rs 7000, Rs 9500

(E) None of these

Q19 The income of A, B, and C are in the ratio of 3:5:7 and their expenditures are in the ratio of 5:7:9. If A saves 66.66% of his income. Find the ratio of C savings to that of B savings.

- (A) 26:19 (B) 17:24
(C) 13:9 (D) 24:17
(E) None of these

Q20 The students in 3 classes are in the ratio 3:4:5. If 20 students are added in each class, the ratio becomes 5:6:7. Find the total number of students in all the 3 classes now.

- (A) 200 (B) 180
(C) 160 (D) 175
(E) None of these



Level-2

- Q1** Salaries of akash, bablu and chintu are in the ratio of 2:3:5 . If their salaries were increased by 15% , 10% and 20% respectively, then what would be the new ratio of their salaries?
 (A) 3:3:10 (B) 23:33:60
 (C) 20:22:40 (D) 1:2:3
 (E) None of these
- Q2** Two numbers are 40% and 60% more than the third number respectively. Find the ratio of difference between the two numbers and the third number.
 (A) 1: 5 (B) 2 : 3
 (C) 3 : 7 (D) 7 : 2
 (E) 1: 2
- Q3** If the price of a diamond, which costs Rs. 15000 and weighs 70 grams, is determined by the square of its weight and it is cut into two pieces in the ratio of 3 : 7, what is the loss incurred?
 (A) Rs.6600 (B) Rs.6500
 (C) Rs.6400 (D) Rs. 6300
 (E) None of these
- Q4** The ratio of the first and second-class fares between the two stations is 7 : 3 and the number of passengers travelling by first and second-class is 4 : 6. If Rs. 46000 is collected as fare, what is the amount collected from first class passengers?
 (A) Rs.27600 (B) Rs.29900
 (C) Rs.28000 (D) Rs.29000
 (E) None of these
- Q5** The expenditure ratio of A, B and C is 16: 12: 9. If they save 20%, 25% and 40% of their income respectively. If the total income of A, B and C is Rs. 10200. Then find the income of A.
 (A) 2000 (B) 3000
 (C) 4000 (D) 5000
 (E) None of these
- Q6** In the income statement of Ashu and Ravi, the ratio of their income in the year 2019 was 5: 4. The proportion of Ashu's income in the year 2020 to that in 2019 is 3: 5 and the ratio of Ravi's income in the year 2019 to that in 2020 is 2: 3. If the sum of the payment of Ashu and Ravi in the year 2020 is Rs. 10242, then find the income of Ravi in the year 2019.
 (A) Rs. 1024 (B) Rs. 1138
 (C) Rs. 2776 (D) Rs. 2420
 (E) Rs. 4552
- Q7** Meesha went to an ice cream parlor and ordered 2 Cornetto double chocolate frozen desserts for Rs 40 each. She gave the owner a 200 Rs note. The owner gave the balance money in coins of Rs 1, Rs 2, and Rs 5 in the ratio 3:6:5. Find the number of 2 Rs coins she got.
 (A) 24 (B) 12
 (C) 18 (D) 30
 (E) None of these
- Q8** The ratio of income of P to that of Q is 1:3. If P and Q saves Rs 2400 and Rs 6400 respectively and the expenditure of P is equal to 42.85% of the expenditure of Q then the two-third of the total income of P and Q is?
 (A) 4200 (B) 2400
 (C) 4800 (D) 3200
 (E) None of these
- Q9** The ratio of expenditure of A, B and C is 18:20:28 and their total income is Rs 1309. Find the total income of A if they save 14.28%, 16.66% and 12.5% of their incomes?
 (A) Rs 578 (B) Rs 595
 (C) Rs 357 (D) Rs 408
 (E) None of these
- Q10** Marks obtained by Mohit in History(H) and Geography(G) are in the ratio of 4 : 5, respectively, while the ratio of marks obtained by Mohit in Polity(P) and Economics(E) is 5 : 6, respectively. He scored 72 marks more in



Economics than in Geography, and 64 marks more in Polity than in History. Find the ratio of marks obtained by Mohit in History and Economics.

- (A) 9:8 (B) 32:42
(C) 42:32 (D) 8:9
(E) 7:4

Q11 The number of audience in a Cinema Hall falls in the ratio 4:3, when the ticket prices in a cinema hall increases in the ratio 7:8. Find the total money collected by cinema hall at present, if before price increase the collection was Rs 93800.

- (A) Rs 70350 (B) Rs 87100
(C) Rs 80400 (D) Rs 77050
(E) None of these

Q12 In a train, the ratio of male to female in general coaches is 6: 5, ratio of male to female in sleeper coaches is 2: 3 and ratio of male to female in AC coaches is 4: 3. Number of males in general coaches is equal to those in sleeper coaches and also 20 less than those in AC coaches. If the total number of males in the train is 200, then find the ratio of total males to total females in the train?

- (A) 10: 9 (B) 1: 1
(C) 5: 4 (D) 20: 19
(E) 4: 3

Q13 Salaries of two friends P and Q are 8:7 in ratio. If the salary of Q increases by 20% and the salary of P increases by 10% the new ratio becomes 96:77 respectively. What was P's salary?

- (A) 9600
(B) 7700
(C) 800
(D) Can't be determined
(E) None of these

Q14 Ratio between salaries of A, B and C is $\frac{2}{5} : \frac{4}{5} : \frac{1}{2}$ respectively. A spends 50% of his salary, B spends 25% of his salary and the difference between their savings is ₹12000. If C

spends ₹5000, then find the ratio between savings of A, B and C respectively.

- (A) 3: 8: 6 (B) 6: 18: 7
(C) 2: 7: 5 (D) 3: 9: 5
(E) 6: 5: 7

Q15 Rs 20790 is divided among Nobita, Zian and Sunio in such a way such that Nobita gets 80% of what Zian gets and Zian gets 66.66% of what Sunio gets. How much more or less does Sunio gets from Zian?

- (A) Rs 4410 (B) Rs 3780
(C) Rs 3150 (D) Rs 3465
(E) None of these

Q16 In an olympiad the ratio of the number of students who scored 99 percentile to that of 90 percentile is 4:25. If 30 more students appeared for the olympiad the number of students scoring 99 percentile increases by 1. If at present the ratio of 99 percentiles to that of 90 percentiles is 22:3. Find the number of students who appeared initially.

- (A) 116 (B) 174
(C) 145 (D) 203
(E) None of these

Q17 Salaries of Nitin, Prem are in the ratio of 5 : 7 and that of Prem and Rahul are in the ratio of 3 : 5. Salary of Nitin is Rs. 82500 and Rahul spends two-seventh of his salary on ration. How much money is left with Rahul after his expenditure on ration?

- (A) 225000 (B) 125000
(C) 137500 (D) 121500
(E) None of these

Q18 Ratio of income of Ahem to that of Vehem is 4:7. If the expenditure of Ahem is 66.66% of his income and that of Vehem is 28.56% of his income and their total savings is Rs 1520. Find the 27.27% of total income of Ahem and Vehem?

- (A) Rs 1173 (B) Rs 1760
(C) Rs 720 (D) Rs 978
(E) None of these



Q19 A sum of ₹($x + 4500$) is distributed among A, B and C such that, A got ₹9y, B got ₹8y and C got ₹12y. If the total amount is 625% more than the difference between the amounts received by B and C, then find the ratio of 'x' to 'y'?

- (A) 20: 1
- (B) Can't be determined
- (C) 15: 1
- (D) 40: 1

(E) 17:1

Q20 The ratio of income of Rahul for two consecutive years is 2:3 and the ratio of his expenditures is 5:9. If Rahul's income for the second year is Rs 45,000 and the expenditure of the first year is Rs 25,000. Then his savings of two years is ?

- (A) Rs 6000
- (B) Rs 5000
- (C) Rs 5500
- (D) Rs 4500
- (E) None of these



Level-3

- Q1** Three friends P, Q and R collected an amount of Rs 9600 by saving each day. Each one of them has contributed different amount each day as per they received from their parents. They distributed among themselves in the ratio 7:8:9. If they spend in the ratio 3:2:4 and R spends Rs 400. Find the ratio of savings of P, Q and R?
 (A) 24:31:33 (B) 7:8:9
 (C) 25:30:32 (D) 4:5:6
 (E) None of these
- Q2** Ratio of total people of an area who read newspaper A to newspaper B is 2: 3. Ratio of total people who read Hindi newspaper A to English newspaper A is 4: 1 and ratio of total people who read Hindi newspaper B to English newspaper B is 1: 2. Find the ratio of total people who read Hindi newspaper (both A and B) to who read English newspaper (both A and B).
 (A) 13: 12 (B) 7: 6
 (C) 9: 8 (D) 4: 3
 (E) 7:9
- Q3** Some 2 rupee coins and 5 rupee coins are distributed between A and B such that, A and B received 2 rupee coins in the ratio of 5: 4 and 5 rupee coins in the ratio of 2: 3 respectively. If A received a total ₹80 and the amount received by B in the form of 5 rupee coins is ₹51 more than that received in the form of 2 rupee coins, then find the total amount received by B.
 (A) ₹90 (B) ₹105
 (C) ₹99 (D) ₹93
 (E) ₹102
- Q4** The total number of coins in a box is 500 and the number of coins made of steel are mixed with the number of coins made of iron in the box and their ratio is 3:2. If 33.33% of Steel made coins are converted into iron coins, then find the total number of iron made coins.
 (A) 300 (B) 200
- (C) 500 (D) 600
 (E) 700
- Q5** The ratio of weight of soaked gram seeds to that of soaked kidney beans is 8:9 and the total weight of both is 170gm. The seeds are dried in the sunlight to make fried dals. The weight of gram seeds is reduced by 8gm and the new ratio of weight becomes 6:5. Find the amount by which the weight of kidney beans is reduced.
 (A) 12gm (B) 24gm
 (C) 30gm (D) 18gm
 (E) None of these
- Q6** The ratio of last year income of P ,Q and R is 3:4:5 and the ratio of their last year income to current year income is 4:5 , 2:3 , and 3:4 respectively. If their total income is Rs 88,650, Find the value of x if P saves Rs 12960 after spending x% of her total income.
 (A) 64 (B) 72
 (C) 36 (D) 42
 (E) None of these
- Q7** The ratio of the cement, rock and sand in making a house is 2:3:4 and the total quantity of material used in the making of the house is 180 kg. If the amount of cement used is (x+20) kg and the amount of rock used in making another house is x% more than the previous quantity of rock used, then find the quantity of rock used in making the new house.
 (A) 70 (B) 75
 (C) 72 (D) 80
 (E) 90
- Q8** The sum of the monthly incomes of six friends (Amit, Chandan, Divakar, Ganesh, Hitesh and Rohit) is Rs 306000. The income of Chandan, Divakar, and Hitesh are in the ratio of 2: 3: 1. The average of monthly income of Amit and Rohit is Rs 45000. The monthly income of Divakar and



Ganesh is in the ratio of 5: 2. Monthly income of Amit is Rs 45000 less than the monthly income of Chandan. Find the respective ratio between the total monthly income of Chandan and Hitesh and the total monthly income of Divakar and Rohit.

- (A) 6: 11 (B) 7: 13
 (C) 5: 13 (D) 7: 17
 (E) None of these

Q9 The ratio of cost of economy class to that of business class flight ticket between Delhi to Varanasi is 11:9. During the festival season the cost of both types of flight tickets increases in the ratio 4:5 and 6:7 respectively. If the number of passengers travelling during the festive season in both types is 5:7 and the total revenue obtained is Rs 1,13,800. Find the total revenue generated by business class tickets?

- (A) Rs 57,800 (B) Rs 53,000
 (C) Rs 58,800 (D) Rs 55,000
 (E) None of these

Q10 A company manager distributed bonus of Rs 121,000 to her employees A, B and C for their extraordinary work dedication for the company in the ratio $3^{6\frac{1}{3}} : 9^{4\frac{1}{6}} : 27^{2\frac{1}{9}}$. The employee who got the highest bonus spent 35% of his bonus for buying a new bike, 25% for his daughter's coaching, 15% for his wife's saree and the rest he invested in a scheme at a compound interest of 20% p.a . Find the amount he got after 2 years from his investment?

- (A) Rs 44,550 (B) Rs 35,640
 (C) Rs 24,750 (D) Rs 34,650
 (E) None of these



Answer Key

Level-1

Q1 (B)

Q2 (A)

Q3 (B)

Q4 (E)

Q5 (D)

Q6 (B)

Q7 (C)

Q8 (E)

Q9 (A)

Q10 (C)

Q11 (C)

Q12 (C)

Q13 (C)

Q14 (C)

Q15 (C)

Q16 (A)

Q17 (A)

Q18 (A)

Q19 (C)

Q20 (B)



Level-2

Q1 (B)

Q2 (A)

Q3 (D)

Q4 (C)

Q5 (C)

Q6 (E)

Q7 (C)

Q8 (D)

Q9 (C)

Q10 (D)

Q11 (C)

Q12 (B)

Q13 (D)

Q14 (D)

Q15 (C)

Q16 (C)

Q17 (C)

Q18 (C)

Q19 (B)

Q20 (B)



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Level-3

Q1 (C)

Q2 (A)

Q3 (C)

Q4 (A)

Q5 (C)

Q6 (C)

Q7 (C)

Q8 (A)

Q9 (C)

Q10 (B)



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Hints & Solutions

Level-1

Q1 Text Solution:

$$\text{Given, } p = \frac{1}{3}q \Rightarrow \frac{p}{q} = \frac{1}{3} \dots (1)$$

$$\text{And, } q = \frac{1}{2}r \Rightarrow \frac{q}{r} = \frac{1}{2} \text{ or } \frac{q}{r} = \frac{3}{6} \text{ (multiply by 3 to make q equal) } \dots (2)$$

So, Comparing (1) & (2) p:q:r = 1:3:6

Q2 Text Solution:

$$30\% \text{ of } (a + b) = 40\% \text{ of } (a - b)$$

$$\Rightarrow \frac{30}{100}(a + b) = \frac{40}{100}(a - b)$$

$$\Rightarrow 3(a + b) = 4(a - b)$$

$$\Rightarrow 3a + 3b = 4a - 4b$$

$$\Rightarrow a = 7b$$

$$\Rightarrow \frac{a}{b} = \frac{7}{1}$$

$$\Rightarrow a:b = 7:1$$

Q3 Text Solution:

Let the fourth Proportional of 36, 48 and 72 is x

$$\text{Now, } \frac{36}{48} = \frac{72}{x}$$

$$\Rightarrow x \times 36 = 72 \times 48$$

$$\Rightarrow x = 96$$

So the fourth Proportional of 36, 48 and 72 is 96

Q4 Text Solution:

$$A = 2B$$

$$A : B = 2 : 1$$

$$C = 2B$$

$$C : B = 2 : 1$$

$$B : C = 1 : 2$$

$$A : B : C = 2 : 1 : 2$$

Hence the correct answer is "E"

Q5 Text Solution:

$$\text{Consider } 4A = 5B = 6C = k$$

$$A = \frac{k}{4}, B = \frac{k}{5}, C = \frac{k}{6}$$

$$A : B : C = \frac{k}{4} : \frac{k}{5} : \frac{k}{6}$$

LCM of 4, 5 and 6 is 60. Consider this as k A : B :

$$C = \frac{60}{4} : \frac{60}{5} : \frac{60}{6} = 15 : 12 : 10$$

Hence the correct answer is "D"

Q6 Text Solution:

$$\Rightarrow \frac{5x + 2y}{4x + 9y} = \frac{21}{13}$$

$$\Rightarrow 65x + 26y = 84x + 189y$$

$$\Rightarrow 19x = 163y$$

$$\Rightarrow \frac{x}{y} = \frac{163}{19}$$

$$\Rightarrow x:y = 163:19$$

Q7 Text Solution:

Let the number of red balls and blue balls in the bag is 3x and 5x respectively.

Then, according to question

$$3x + 5x = 24$$

$$8x = 24$$

$$x = 3$$

Number of red balls in the bag = 3x = 3 × 3 = 9 balls

Q8 Text Solution:

Let the number to be subtracted is 'x'

$$\text{Then, } \frac{17-x}{23-x} = \frac{27-x}{38-x}$$

$$\Rightarrow (17-x)(38-x) = (27-x)(23-x)$$

$$\Rightarrow 646 - 17x - 38x + x^2 = 621 - 27x$$

$$- 23x + x^2$$

$$\Rightarrow 646 - 55x = 621 - 50x$$

$$\Rightarrow 25 = 5x$$

$$\Rightarrow 5 = x$$

number to be subtracted is 5 and it makes the numbers in proportion

$$\Rightarrow \frac{17-x}{23-x} = \frac{27-x}{38-x}$$

$$\Rightarrow \frac{17-5}{23-5} = \frac{27-5}{38-5}$$

$$\Rightarrow \frac{12}{18} = \frac{22}{33}$$

$$\Rightarrow \frac{2}{3} = \frac{2}{3}$$

Q9 Text Solution:

According to the question,

$$\text{Required sum} = \frac{(5 + 7 + 9)}{9} \times 6300$$

$$= \text{Rs. } 14,700$$

Hence, option a.

Q10 Text Solution:

Let the three numbers be A, B and C.

Then according to question,

$$A : B = 2:3 \dots (i)$$

$$B : C = 5:8 \dots (ii)$$

To make ratio of B equal in both equations multiply (i) by 5 and (ii) by 3

$$A : B = 10 : 15$$



$$B : C = 15 : 24$$

$$\Rightarrow A : B : C = 10 : 15 : 24$$

$$\text{Given, } 10x + 15x + 24x = 1176$$

$$49x = 1176$$

$$x = \frac{1176}{49}$$

$$x = 24$$

$$\text{Third number} = 24x$$

$$= 24 \times 24 = 576$$

Q11 Text Solution:

$$\text{Original price/price after increase} = \frac{9}{11}$$

$$18000/\text{price after increase} = \frac{9}{11}$$

$$\text{price after increase} = 22,000$$

$$\text{Price increase} = 22000 - 18000 = 4000$$

Q12 Text Solution:

Let the income of Riya and Supriya be $5x$ and $7x$

let the expenditure of Riya and Supriya be $7y$ and $10y$

According to the question,

$$5x = 10y \Rightarrow x = 2y.$$

Ratio of saving of Riya and Supriya

$$= (5x - 7y) : (7x - 10y)$$

$$= (10y - 7y) : (14y - 10y)$$

$$= 3 : 4$$

Q13 Text Solution:

Given:

The ratio of the price of a bed and a table is $9 : 7$

The average price of two beds and a table is Rs. 60,000

Formula Used:

$$\text{Average} = \frac{\text{sum of all the observations}}{\text{total number of observations}}$$

Explanation :

Let the price of a bed and a table be Rs. $9M$ and Rs. $7M$ respectively.

According to the question,

The average price of two beds and a table is Rs. 60,000

$$\frac{2 \times 9M + 7M}{3} = 60,000$$

$$18M + 7M = 1,80,000$$

$$M = 7,200$$

$$\text{Price of a bed} = 9 \times 7,200 = \text{Rs. } 64,800$$

$$\text{Price of a table} = 7 \times 7,200 = \text{Rs. } 50,400$$

$$\therefore \text{Required sum} = 64,800 + 50,400 = \text{Rs. } 1,15,200$$

Hence, "Rs. 1,15,200" is the correct answer.

Q14 Text Solution:

The ratio of three numbers is $3:5:9$

Let the numbers be $3x$, $5x$ and $9x$

Their squares will be $9x^2$, $25x^2$ and $81x^2$

Sum of their squares is

$$9x^2 + 25x^2 + 81x^2 = 4140$$

$$\Rightarrow 115x^2 = 4140$$

$$\Rightarrow x^2 = 36$$

$$\Rightarrow x = 6$$

Sum of the numbers is $3x + 5x + 9x = 17x$

$$\Rightarrow 17x = 17 \times 6$$

$$\Rightarrow 102$$

Q15 Text Solution:

Let the total number of boys in the school be x .

Total number of girls in the school = $x - 360$

According to the question,

$$x + (x - 360) = 1680$$

$$2x = 1680 + 360$$

$$x = \frac{2040}{2} = 1020$$

$$\text{Number of girls} = 1020 - 360 = 660$$

Required ratio = $1020 : 660$

$$= \frac{1020}{660} = 17 : 11$$

Option '17:11' is the correct answer

Q16 Text Solution:

The ratio of three numbers A, B and C is $13:17:23$ and their sum is 265

$$\text{Let } A = 13x$$

$$B = 17x$$

$$C = 23x$$

$$\text{And } 13x + 17x + 23x = 265$$

$$53x = 265$$

$$x = 5$$

$$\text{So } A = 13x \Rightarrow 65$$

$$B = 17x \Rightarrow 85$$

$$C = 23x \Rightarrow 115$$

ratio between the sum of A and B to the difference of C and B

$$\Rightarrow \frac{A+B}{C-B}$$

$$\Rightarrow \frac{65+85}{115-85}$$

$$\Rightarrow \frac{5}{1}$$

Q17 Text Solution:

Let the income of M = Rs. $8x$

Income of N = Rs. $11x$

According to the question,

$$(8x - 6000):(11x - 6000) = 5:8$$

$$64x - 48000 = 55x - 30000$$

$$9x = 18000$$

$$x = 2000$$

Income of M = Rs. 16000

Q18 Text Solution:

According to the question,

1 unit = 2500.

So, the income of Utkarsh
 $= Rs\ 4 \times 2500 = Rs\ 10000$

And the income of Radhika
 $= Rs\ 3 \times 2500 = Rs\ 7500$

Q19 Text Solution:

Let the incomes of A, B and C are $3x$, $5x$, $7x$.

And their expenditures are $5y$, $7y$, $9y$ respectively.

ATQ, A saving's = 66.66% of A's income

$$\Rightarrow (3x - 5y) = \frac{2}{3} \times 3x$$

$$\Rightarrow (3x - 5y) = 2x$$

$$\Rightarrow (3x - 2x) = 5y$$

$$\Rightarrow x = 5y$$

So, the ratio of C saving's to that of B saving's

$$= (7x - 9y) : (5x - 7y)$$

$$= (35y - 9y) : (25y - 7y) := 26y : 18y$$

$$= 13 : 9.$$

Q20 Text Solution:

According to the question,

$$\frac{3x+20}{4x+20} = \frac{5}{6}$$

$$18x + 120 = 20x + 100$$

$$2x = 20$$

$$x = 10$$

Required total number of students = $(5 + 6 + 7) \times$

$$10 = 180$$



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Level-2

Q1 Text Solution:

Let the salaries of Akash, Bablu, and Chintu will be $2x$, $3x$, and $5x$ respectively.

After 15% increase in Akash's salary $= 15 \times \frac{2x}{100} = 0.3x$

New salary $= 2x + 0.3x = 2.3x$

New salary of Bablu $= 3x + 10 \times \frac{3x}{100} = 3.3x$

New salary of Chintu $= 5x + 20 \times \frac{5x}{100} = 6x$

New ratio $= 2.3x : 3.3x : 6x = 23:33:60$

Q2 Text Solution:

Let the third number be x

Then first number will be $\frac{140x}{100} = \frac{7x}{5}$

And second number will be $\frac{160x}{100} = \frac{8x}{5}$

Difference of two numbers $= \frac{8x}{5} - \frac{7x}{5} = \frac{x}{5}$

Required ratio $= \frac{\frac{x}{5}}{\frac{x}{1}} = \frac{x}{5} \times \frac{1}{x}$

$= 1 : 5$

Q3 Text Solution:

New ratio is $3 : 7$.

The price of the diamond before broken is Rs. 15000.

Let the weight of each piece of diamond be $3x$ and $7x$.

Total weight of diamond $= 3x + 7x = 10x$

Cost of the diamond $= (10x)^2 = 100x^2$

Cost of first piece $= (3x)^2 = 9x^2$

Cost of second piece $= (7x)^2 = 49x^2$

Total cost of diamond after weight $= 9x^2 + 49x^2 = 58x^2$

$\Rightarrow$ New cost $= 58x^2$

According to the question,

$100x^2 = 15000$

$\Rightarrow x^2 = 150$

The loss in the price of diamond after broken $= 100x^2 - 58x^2$

$\Rightarrow$ The required loss $= 42x^2$

$\Rightarrow 150 \times 42$

$\Rightarrow$ Rs.6300

$\therefore$ The loss in the price of the diamond after breaking is Rs.6300.

Q4 Text Solution:

Ratio of amount collected from first and second class $= 7 \times 4 : 3 \times 6$

$\Rightarrow 28 : 18 = 14 : 9$

Let amount collected from first and second class be $14x$ and $9x$ respectively,

According to question,

$14x + 9x = 46000$

$23x = 46000$

$x = 2000$

Amount collected from first class passengers $= 14 \times 2000$

$\Rightarrow$ Rs.28000

Q5 Text Solution:

Let income of A, B and C be x , y and z respectively.

$20\% = \frac{1}{5}$, $25\% = \frac{1}{4}$ and $40\% = \frac{2}{5}$

Hence, savings of A, B and C will be $\frac{x}{5}$, $\frac{y}{4}$ and $\frac{2z}{5}$.

Now, expenditure of A $= x - \frac{x}{5} = \frac{4x}{5}$

Expenditure of B $= y - \frac{y}{4} = \frac{3y}{4}$

Expenditure of C $= z - \frac{2z}{5} = \frac{3z}{5}$

According to question

$\frac{4x}{5} : \frac{3y}{4} = 16 : 12$

$\frac{16x}{15y} = 16 : 12$

$\Rightarrow x : y = 5 : 4$ ----(1)

And, $\frac{3y}{4} : \frac{3z}{5} = 12 : 9$

$\frac{3y}{4} : \frac{3z}{5} = 4 : 3$

$y : z = 16 : 15$ ----(2)

Combining (1) and (2), we get

$x : y : z = 20 : 16 : 15$

Given That total income of A, B and C are $= 10200$

$\therefore$ Income of A $= \left(\frac{20}{51}\right) \times 10200 = \text{Rs. } 4000$.

Q6 Text Solution:

Let the income of Ashu in 2020 and 2019 be $3x$ and $5x$ respectively.

Let the income of Ravi in 2020 and 2019 be $3y$ and $2y$ respectively

Since, the ratio of their income in the year 2019 was $5 : 4$



$$\Rightarrow 5x : 2y = 5 : 4$$

$$\Rightarrow 2x = y$$

The sum of their incomes in 2020 is Rs. 10242

$$\Rightarrow 3x + 3y = 10,242$$

$$\Rightarrow 9x = 10,242$$

$$\Rightarrow x = 1,138 \text{ and } y = 2276$$

Ravi's income for the year 2019 = $2y$ = Rs. 4552

Q7 Text Solution:

Total amount of 2 Cornetto
= $Rs(2 \times 40) = Rs80$

Amount left with Meesha
= $Rs(200 - 80) = Rs 120$.

Ratio of Rs1, Rs2 and Rs5 = $3 : 6 : 5$

Ratio of amount of coins = $3 \times 1 : 6 \times 2 : 5 \times 5$
= $3 : 12 : 25$

ATQ, $(3 + 12 + 25) \text{ units} = 120$

$(40) \text{ units} = 120 \Rightarrow 1 \text{ unit} = 3$

So, the number of Rs 2 coins = $6 \times 3 = 18$

Q8 Text Solution:

Let the income of P and Q be x and $3x$ respectively.

ATQ, $\frac{x-2400}{3x-6400} = \frac{3}{7} \text{ use } (42.85\% = \frac{3}{7})$

$$7 \times (x - 2400) = 3 \times (3x - 6400)$$

$$7x - 16800 = 9x - 19200$$

$$19200 - 16800 = 9x - 7x$$

$$2400 = 2x \Rightarrow x = 1200.$$

So, the total income of P and Q is
= $(x + 3x) = 4x = 4 \times 1200 = 4800$.

So, two-third of total income
= $\frac{2}{3} \times 4800 = 3200$.

Q9 Text Solution:

Let the income of A, B and C be x, y, z .

$$\text{Using } 14.28\% = \frac{1}{7}$$

$$16.66\% = \frac{1}{6}$$

$$12.5\% = \frac{1}{8}$$

Then the expenditure of A, B and C are in the ratio

$$= (1 - \frac{1}{7})x : (1 - \frac{1}{6})y : (1 - \frac{1}{8})z$$

$$= (\frac{6}{7})x : (\frac{5}{6})y : (\frac{7}{8})z \dots\dots\dots(1)$$

ATQ, Ratio of expenditure of A, B and C is
18:20:28...(2)

On comparing 1 and 2, $x = 21, y = 24, z = 32$.

Also, total income = 1309.

$$\text{So Income of A} = \frac{21}{(21+24+32)} \times 1309 = 357$$

Q10 Text Solution:

Given, H:G = 4:5, P:E = 5:6

To find, H:E

Let, Mohit's marks in H be $4x$ & in G be $5x$

And, Mohit's marks in P be $5y$ & in E be $6y$

According to question, $E - G = 6y - 5x = 72$ & $P-H$
= $5y - 4x = 64$

$$\text{So, } 6y - 5x = 72 : 5y - 4x = 64$$

$$\text{Thus, } X = 32 ; Y = 24$$

$$\text{So, } H = 4x = 4(32) = 128 \text{ \& } E = 6y = 144$$

$$\text{Hence, } H:E = 128:144 = 8:9$$

Q11 Text Solution:

Let the initial number of audience be $4x$

Then the number of audience after the price increase = $3x$

Also, let the initial ticket price = $7y$

And ticket price after increase = $8y$

Total amount collected = Ticket price $\times$ Number of audience

$$\text{ATQ, } (4x \times 7y) = 93800 \Rightarrow xy = 3350.$$

Hence total amount collection after price increase = $(3x \times 8y)$

$$= (24xy)$$

$$= (24 \times 3350)$$

$$= 80400$$

Q12 Text Solution:

Let the number of males in general coaches is $6x$.

So, the number of males in sleeper coaches = $6x$

And the number of males in AC coaches = $(6x + 20)$

From the question:

$$6x + 6x + 6x + 20 = 200$$

$$x = 10$$

Number of males in general coaches = $6 \times 10 = 60$

Number of males in sleeper coaches = 60

Number of males in AC coaches = $60 + 20 = 80$

Number of females in general coaches =
 $60 \times \frac{5}{6} = 50$



Number of females in sleeper coaches = $60 \times \frac{3}{2} = 90$
 Number of females in AC coaches = $80 \times \frac{3}{4} = 60$
 Total number of females in the train = $50 + 90 + 60 = 200$
 Required ratio = $200 : 200 = 1 : 1$

Q13 Text Solution:

Let, Salary of P be 80, and Q be 70
 20% increase in P = 120% of 80 = $\frac{120}{100} \times 80 = 96$
 10% increase in Q = 110% of 70 = $\frac{110}{100} \times 70 = 77$
 Hence, Can't be determined, as no value is provided for the ratio.

Q14 Text Solution:

Ratio between salaries of A, B and C = $\frac{2}{5} : \frac{4}{5} : \frac{1}{2} = 4 : 8 : 5$
 Let the salaries of A, B, and C be ₹4x, ₹8x and ₹5x respectively.
 So, A's savings = 50% of 4x = ₹2x
 And B's savings = 75% of 8x = ₹6x
 From the question:
 $6x - 2x = 12000$
 $x = 3000$
 A's salary = $4 \times 3000 = ₹12000$
 A's savings = 50% of 12000 = ₹6000
 B's salary = $8 \times 3000 = ₹24000$
 B's savings = 75% of 24000 = ₹18000
 C's salary = $5 \times 3000 = ₹15000$
 C's savings = $15000 - 5000 = ₹10000$
 Required ratio = $6000 : 18000 : 10000 = 3 : 9 : 5$

Q15 Text Solution:

Nobita gets 80% of what Zian gets
 $\Rightarrow \text{Nobita} : \text{Zian} = 4 : 5$
 Zian gets 66.66% of what Sunio gets
 $\Rightarrow \text{Zian} : \text{Sunio} = 2 : 3$
 Equivalent ratio of Nobita, Zian and Sunio
 $= 8 : 10 : 15$
 The amount Sunio gets more of Zian
 $= \frac{(15-10)}{33} \times 20790 = 3150$

Q16 Text Solution:

Let the number of 99 percentilers = $4x$
 and the number of 90 percentilers = $25x$
 Total students initially = $29x$
 ATQ, $\frac{29x+30}{4x+1} = \frac{(22+3)}{3}$
 $(29x + 30) \times 3 = (4x + 1) \times 25$
 $87x + 90 = 100x + 25 \Rightarrow x = 5$
 Total number of students initially
 $= 29 \times 5 = 145$

Q17 Text Solution:

Nitin's Salary : Prem's salary = $5 : 7$ and Prem's salary : Rahul's salary = $3 : 5$
 Nitin's salary : Prem's salary : Rahul's salary
 $= 15 : 21 : 35$
 Salary of Rahul = $\frac{35}{15} \times 82500 = 192500$
 Rahul's expenditure on ration
 $= \frac{2}{7} \times 192500 = \text{Rs. } 55000$
 Money left with Rahul
 $= 192500 - 55000 = \text{Rs. } 137500$
 Hence Option 'Rs. 137500' is the correct answer.

Q18 Text Solution:

Let income of Ahem = $4x$
 So, Expenditure of Ahem
 $= 66.66\% \text{ of income} = \frac{2}{3} \times 4x = \frac{8x}{3}$
 And savings of Ahem = $4x - \frac{8x}{3} = \frac{4x}{3}$
 Let income of Vehem = $7x$
 So, Expenditure of Vehem
 $= 28.56\% \text{ of income} = \frac{2}{7} \times 7x = 2x$
 And savings of Vehem = $7x - 2x = 5x$
 Total Savings = $\frac{4x}{3} + 5x = \frac{19x}{3}$
 ATQ, $\frac{19x}{3} = 1520 \Rightarrow x = 240$
 Total income of Ahem and Vehem
 $= (4 + 7)x = 11 \times 240 = 2640$
 Hence
 $27.27\% \text{ of total income} = \frac{3}{11} \times 2640$
 $= \text{Rs } 720$

Q19 Text Solution:

Since, the amounts received by A, B and C are ₹9y, ₹8y and ₹12y respectively.
 So,
 $9y + 8y + 12y = x + 4500$
 $29y = x + 4500 \text{ -----(1)}$



Since, the total amount is 625% more than the difference between the amounts received by B and C.

So,

$$x + 4500 = \frac{725}{100} \times (12y - 8y)$$

$$x + 4500 = 29y \text{ -----(2)}$$

Since, both the equations can't be solved.

So, the ratio of 'x' to 'y' can't be determined.

Q20 Text Solution:

Let the income of Rahul for the first and second year be $2x$ and $3x$ respectively.

And the expenditures are $5y$ and $9y$ respectively.

$$\text{ATQ, } 3x = 45000 \Rightarrow x = 15000.$$

$$\begin{array}{l} \text{Total income of Rahul} \\ = (2x + 3x) = 5 \times 15000 = 75000 \end{array}$$

$$\text{Also, } 5y = 25000 \Rightarrow y = 5000$$

$$\begin{array}{l} \text{Total expenditure of Rahul} \\ = (5y + 9y) = 14 \times 5000 = 70000 \end{array}$$

$$\begin{array}{l} \text{Total savings of Rahul} \\ = 75000 - 70000 = 5000. \end{array}$$



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Level-3

Q1 Text Solution:

Let the distribution of amounts among P, Q, and R be $7x$, $8x$, and $9x$ respectively.

$$\text{ATQ, } (7 + 8 + 9)x = 9600$$

$$24x = 9600$$

$$\text{i.e. } x = 400.$$

So the respective amounts of P, Q, and R are 2800, 3200, and 3600.

Let the expenditures be in the ratio $3y$, $2y$, and $4y$
ATQ, $4y = 400$.

$$y = 100.$$

So the respective expenditures are 300, 200 and 400.

Hence the savings of P, Q, and R are 2500, 3000, 3200.

So the ratio is 25:30:32.

Q2 Text Solution:

Let total people who read newspaper A and B is ' $2x$ ' and ' $3x$ ' respectively.

People who read Hindi newspaper A =

$$2x \times \frac{4}{5} = 1.6x$$

People who read English newspaper A =

$$2x \times \frac{1}{5} = 0.4x$$

People who read Hindi newspaper B =

$$3x \times \frac{1}{3} = x$$

People who read English newspaper B =

$$3x \times \frac{2}{3} = 2x$$

Total people who read Hindi newspaper (both A and B) = $1.6x + x = 2.6x$

Total people who read English newspaper (both A and B) = $0.4x + 2x = 2.4x$

Required ratio = $2.6x : 2.4x = 13 : 12$

Q3 Text Solution:

Let the number of 2 rupees coins received by A and B be $5x$ and $4x$ respectively.

Also, let the number of 5 rupee coins received by A and B be $2y$ and $3y$ respectively.

So,

$$5x \times 2 + 2y \times 5 = 80$$

$$x + y = 8 \text{ -----(1)}$$

And,

$$3y \times 5 - 4x \times 2 = 51$$

$$15y - 8x = 51 \text{ -----(2)}$$

From equations (1) and (2):

$$15y - 8(8 - y) = 51$$

$$15y - 64 + 8y = 51$$

$$y = 5$$

From equation (1):

$$x = 3$$

$$\begin{aligned} \text{So, the total amount received by B} &= (4 \times 3 \times 2) \\ &+ (3 \times 5 \times 5) \\ &= ₹99 \end{aligned}$$

Q4 Text Solution:

$$\text{Number of Steel coins} = \frac{3}{5} \times 500 = 300$$

$$\text{Number of iron coins} = 500 - 300 = 200$$

So; after converting steel coins into iron coins;

$$\text{Number of steel coins left} = \frac{1}{3} \times 300 = 200$$

Thus,

$$\text{Number of iron coins} = 200 + 100 = 300$$

Q5 Text Solution:

Let the weight of soaked gram seeds = $8x$

And the weight of soaked kidney beans = $9x$

$$\text{ATQ } (8x + 9x) = 170 \text{ i.e. } x = 10$$

So the weight of soaked gram seeds = 80

the weight of soaked kidney beans = 90

Let the weight of kidney beans be reduced by x gm

$$\text{ATQ } \frac{(80-8)}{(90-x)} = \frac{6}{5}$$

$$72 \times 5 = 6 \times (90 - x) \Rightarrow x = 30.$$

Q6 Text Solution:

Given the ratio of last year income of P, Q, and R is 3:4:5

The ratio of their last year income to the current year's income is 4:5, 2:3, and 3:4

So, the ratio of their current year income is :

$$\frac{3}{4} \times 5 : \frac{4}{2} \times 3 : \frac{5}{3} \times 4.$$

$$\text{i.e. } 45 : 72 : 80.$$

ATQ, Total income of P, Q, and R = Rs 88,650

$$\text{i.e. } (45 + 72 + 80) \text{ units} = 88650$$



197 units = 88650

1 unit = 450.

So, total income of P = $(45) \times 450 = 20,250$

ATQ, $(100 - x)\%$ of 20,250 = 12960

$$\frac{(100-x)}{100} = \frac{12960}{20250} \Rightarrow x = 36.$$

Q7 Text Solution:

Let the quantity of cement, rock and sand used in making the house is $2x$, $3x$ and $4x$ kg respectively

According to the problem,

total quantity of material used in the making of the house is 180 kg. Therefore,

$$2x + 3x + 4x = 180$$

$$\Rightarrow 9x = 180$$

$$\Rightarrow x = 20$$

So, the quantity of cement, rock and sand used in making the house is 40 kg, 60 kg and 80 kg respectively.

Now, the amount of cement used in making another house is $(x+20)$ kg, which means:

Amount of cement used in the new house = $2(x+20) = 2x + 40$ kg

Let's assume that the quantity of rock used in making the new house is y kg.

According to the problem,

amount of rock used in making the new house is $x\%$ more than the previous quantity of rock used, which means:

$$\Rightarrow y = \frac{(100+x)}{100 \times 3x}$$

$$\Rightarrow y = \frac{(100+x)}{100 \times 60}$$

$$\Rightarrow y = \frac{(100+20)}{100 \times 60} = 72 \text{ kg}$$

Therefore, the quantity of rock used in making the new house is 72 kg.

Q8 Text Solution:

Sum of the monthly incomes of six friends = Rs 306000

Let the monthly income of Chandan, Divakar, and Hitesh be Rs $10x$, Rs $15x$, and Rs $5x$ respectively.

Sum of monthly income of Amit and Rohit = $45000 \times 2 = \text{Rs } 90000$

Monthly income of Ganesh = $45000 \times 2 = \text{Rs } 90000$

The ratio of monthly income of Chandan, Divakar, Ganesh and Hitesh = $10x : 15x : 6x : 5x$

The sum of monthly incomes of Chandan, Divakar, Ganesh, and Hitesh = $306000 - 90000 = \text{Rs } 216000$

$$10x + 15x + 6x + 5x = 216000$$

$$x = \frac{216000}{36} = 6000$$

Monthly income of Chandan = $10x = \text{Rs } 60000$

Monthly income of Divakar = $15x = \text{Rs } 90000$

Monthly income of Ganesh = $6x = \text{Rs } 36000$

Monthly income of Hitesh = $5x = \text{Rs } 30000$

Monthly income of Amit = Monthly income of Chandan - 45000 = $60000 - 45000 = \text{Rs } 15000$

Monthly income of Rohit = $90000 - 15000 = \text{Rs } 75000$

Total monthly income of Chandan and Hitesh = $60000 + 30000 = \text{Rs } 90000$

Total monthly income of Divakar and Rohit = $90000 + 75000 = \text{Rs } 165000$

Required ratio = $90000 : 165000 = 6 : 11$

Q9 Text Solution:

The Ratio of the cost of an economy class flight ticket to that of a business class flight ticket = 11:9

After the increase in the ticket price

The new ratio becomes $= \frac{11}{4} \times 5 : \frac{9}{6} \times 7 = 55 : 42$

The number of passengers travelling are in ratio = 5 : 7.

Ratio of revenue generated = $(55 \times 5) : (42 \times 7) = 275 : 294$.

ATQ, $(275 + 294)$ units = 113800

1 unit = 200.

So, revenue generated by business class flight tickets = $(200 \times 294) = 58,800$.

Q10 Text Solution:

Ratio of bonus distribution among A, B and C

$$= 3^{6\frac{1}{3}} : 9^{4\frac{1}{6}} : 27^{2\frac{1}{9}}$$

$$= 3^{6\frac{1}{3}} : 3^{8\frac{1}{3}} : 3^{6\frac{1}{3}}$$

$$= 1 : 9 : 1.$$

Total Bonus amount = 121000



So, $(1+9+1)$ units = 121000

11 units = 121000

1 unit = 11000

Hence , Bonus amount of A, B and C are 11000, 99000 and 11000 respectively.

Accordingly, B obtained the highest bonus of all.

ATQ,B spent 35% of his bonus for buying a new bike, 25% for his daughter's coaching, 15% for his wife's saree

I.e. Bonus amount left with B = 25% of total bonus = Rs 24,750

Amount obtained by investing in a scheme at a compound interest of 20% p.a
 $= 24750 \times \left(1 + \frac{1}{5}\right)^2 = Rs\ 35,640.$



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